Liquid-vapor transition — Solution

W24 (2024) — English translation

A1 — 1.00 pts

Write down an equation that will allow you to use the simple iteration method to find the value $\frac{T_\mathrm{c}}{L_0}$*, and equations that will allow you to find the values* T_c *and* L_0 *from there.*

$$\begin{cases} \dfrac{L_1}{L_0} = \tanh\left[\dfrac{L_1 T_\mathrm{c}}{L_0 T_1}\right] \\ \dfrac{L_2}{L_0} = \tanh\left[\dfrac{L_2 T_\mathrm{c}}{L_0 T_2}\right] \end{cases} \implies L_1 \tanh\left[\frac{L_2 T_\mathrm{c}}{L_0 T_2}\right] = L_2 \tanh\left[\frac{L_1 T_\mathrm{c}}{L_0 T_1}\right]$$

$$\implies \frac{T_\mathrm{c}}{L_0} = \frac{T_2}{L_2} \operatorname{artanh}\left[\frac{L_2}{L_1} \tanh\left[\frac{L_1}{T_1}\frac{T_\mathrm{c}}{L_0}\right]\right]$$

$$\implies L_0 = L_1 \Big/ \tanh\left[\frac{L_1}{T_1}\frac{T_\mathrm{c}}{L_0}\right], \quad T_\mathrm{c} = L_0 \times \frac{T_\mathrm{c}}{L_0}$$

Answer:

$$\frac{T_\mathrm{c}}{L_0} = \frac{T_2}{L_2} \operatorname{artanh}\left[\frac{L_2}{L_1} \tanh\left[\frac{L_1}{T_1}\frac{T_\mathrm{c}}{L_0}\right]\right], \quad L_0 = L_1 \Big/ \tanh\left[\frac{L_1}{T_1}\frac{T_\mathrm{c}}{L_0}\right], \quad T_\mathrm{c} = L_0 \times \frac{T_\mathrm{c}}{L_0}$$

A2 — 4.00 pts

For each pair of points, find the values T_c *and* L_0*.*

Answer:

t_1, °C	L_1, $\frac{\text{kJ}}{\text{kg}}$	t_2, °C	L_2, $\frac{\text{kJ}}{\text{kg}}$	T_c, K	L_0, $\frac{\text{kJ}}{\text{kg}}$
151.8	2108	247.3	1728	652.5	2419
158.8	2086	253.2	1697	653.2	2417
165.0	2066	258.8	1667	653.8	2414
170.4	2048	263.9	1638	654.1	2412
175.4	2030	266.4	1624	654.8	2409
179.9	2014	271.1	1596	655.0	2407
184.1	1999	277.7	1556	655.5	2405
188.0	1985	281.9	1530	655.9	2404
191.6	1971	285.8	1504	656.0	2402
195.0	1958	289.6	1478	656.1	2401
198.3	1946	293.2	1453	656.1	2402
201.4	1934	296.7	1428	656.1	2401
204.3	1922	300.1	1403	656.1	2400
209.8	1900	303.3	1378	656.0	2400
212.4	1889	306.5	1353	656.2	2399
217.2	1868	309.5	1328	656.2	2398
221.8	1849	312.4	1304	656.0	2400
228.1	1820	315.3	1279	656.1	2398
233.8	1794	318.0	1254	655.8	2400
240.9	1760	320.7	1229	655.6	2401

A3 — 0.50 pts

Find the arithmetic average of the values you obtained, $\overline{T}_c$ and $\overline{L}_0$.

Answer:

$$\overline{T}_c = 655.3\ \text{K}, \quad \overline{L}_0 = 2404\ \frac{\text{kJ}}{\text{kg}}$$

B1 — 0.30 pts

Write down a formula that allows you to use the simple iteration method to find $L\left(\frac{1}{T}\right)$ using the values $\overline{L}_0$ and $\overline{T}_c$ obtained in the previous part.

Answer:

$$L = \overline{L}_0 \tanh\left[\frac{\overline{T}_c}{\overline{L}_0} L \frac{1}{T}\right]$$

B2 — 2.00 pts

Using this formula, find the dependence of $L\left(\frac{1}{T}\right)$ in the temperature range from $t = 100$ °C to T_c. Note: Count at least 20 points, try to cover the range uniformly. This will increase the accuracy of your further calculations.

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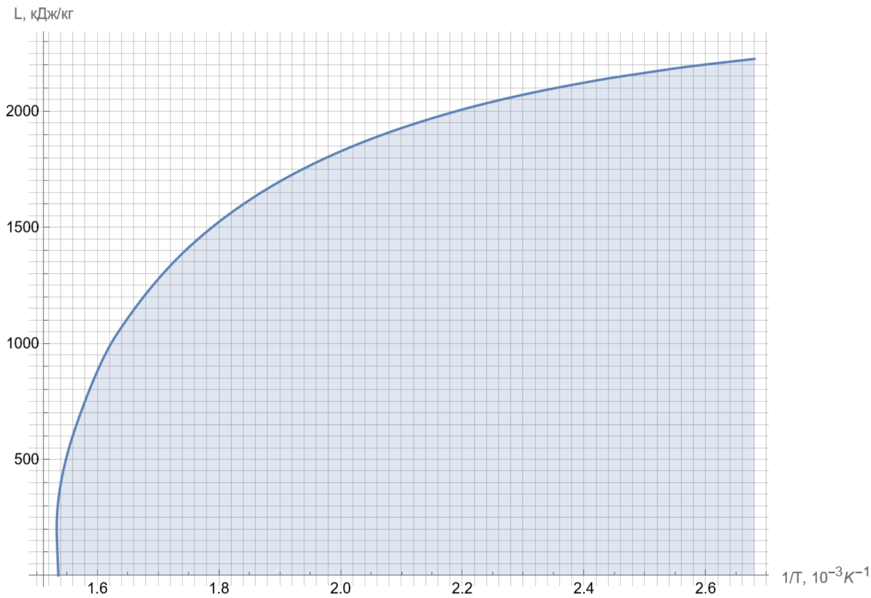
$1/T,\ 10^{-3}\ \mathrm{K}^{-1}$	$L,\ \mathrm{kJ/kg}$	$1/T,\ 10^{-3}\ \mathrm{K}^{-1}$	$L,\ \mathrm{kJ/kg}$
1.54	395	2.14	1961
1.60	879	2.20	2007
1.66	1144	2.26	2047
1.72	1334	2.32	2082
1.78	1481	2.38	2113
1.84	1599	2.44	2141
1.90	1697	2.50	2165
1.96	1779	2.56	2188
2.02	1849	2.62	2207
2.08	1909	2.68	2225

B3 — 0.50 pts

Plot $L\left(\frac{1}{T}\right)$ *from* $t = 100\ ^\circ\mathrm{C}$ *to* T_c.



B4 — 1.00 pts

Calculate the area S *under the graph.*

Answer:

$$S = 2.04 \cdot 10^3\ \frac{\mathrm{J}}{\mathrm{kg} \cdot \mathrm{K}}$$

Note: The area can be calculated using the trapezoidal method, bypassing the direct construction of a graph.

B5 — 0.70 pts

Express the pressure p_c *at the critical point in terms of* S *and calculate it to three significant figures if the pressure is* $p_{sat}(100\ ^\circ\mathrm{C}) = 1.013 \cdot 10^5$ *Pa.*

Answer:

$$p_c = p_{\mathrm{sat}}(100\ ^\circ\mathrm{C}) \exp\left[\frac{\mu S}{R}\right] = 8.36 \cdot 10^6\ \mathrm{Pa}$$

Note: the result turned out to be noticeably underestimated, since the Clapeyron–Clausius equation was written neglecting the specific volume of the liquid phase and the difference between vapor and an ideal gas, which is especially incorrect near the critical point.